

Fitting logistic regression models

Last time: Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Answer: An increase of 1 point on the MCAT is associated with an increase in the odds of acceptance by a factor of $\exp\{0.246\} = 1.28$.

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the estimated probability that a student with an MCAT score of 40 is accepted?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the estimated probability that a student with an MCAT score of 40 is accepted?

$$\text{Answer: } \hat{\pi} = \frac{\exp\{-8.712 + 0.246 \cdot 40\}}{1 + \exp\{-8.712 + 0.246 \cdot 40\}} = 0.76$$

So we estimate that a student with an MCAT score of 40 has a 76% chance of being accepted to medical school.

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Answer: We need odds = 1, so

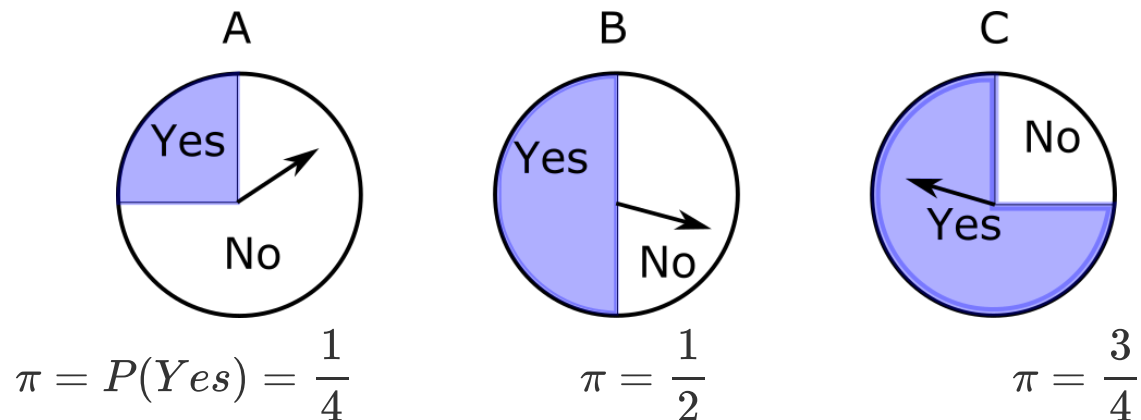
$$\frac{\hat{\pi}}{1 - \hat{\pi}} = \exp\{-8.712 + 0.246 \text{ MCAT}\} = 1$$

So $-8.712 + 0.246 \text{ MCAT} = 0$, which means
 $\text{MCAT} = 8.712/0.246 = 35.4$

Warmup

Work on the warmup activity (handout).

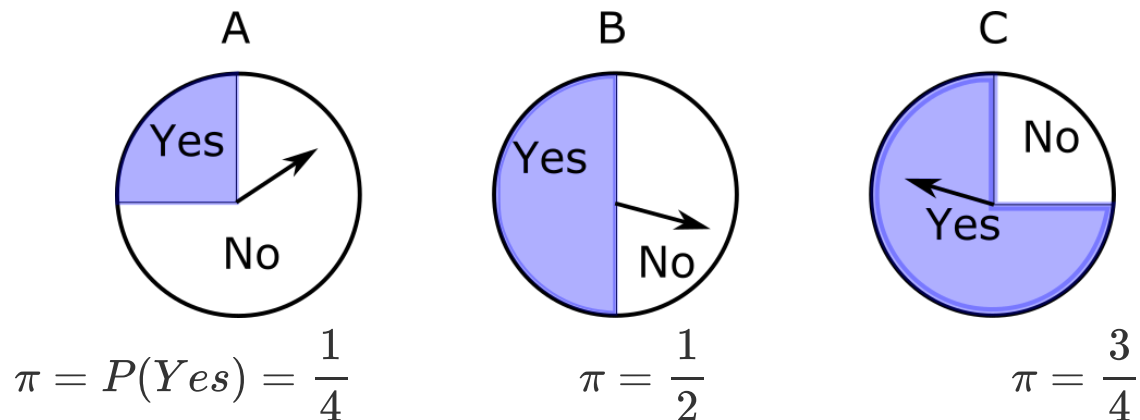
Warmup



A spinner is chosen and spun twice, resulting in No, No.

- + $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$:
- + $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$:
- + $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$:

Warmup



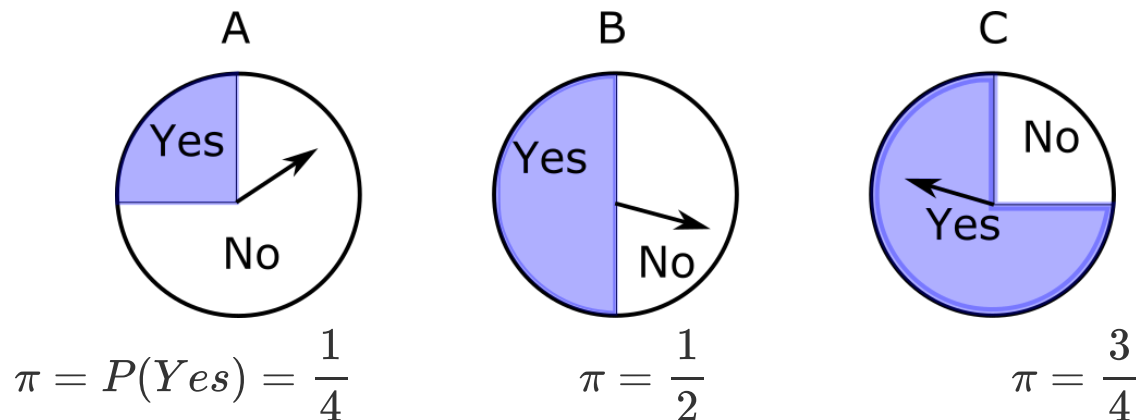
A spinner is chosen and spun twice, resulting in No, No.

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{16}$

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$:

+ $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$:

Warmup



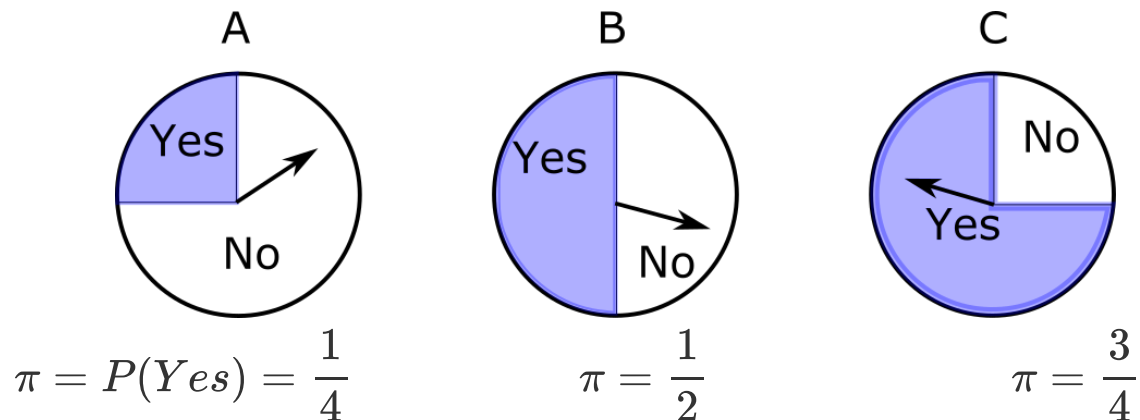
A spinner is chosen and spun twice, resulting in No, No.

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{16}$

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$: $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{4}{16}$

+ $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$:

Warmup



A spinner is chosen and spun twice, resulting in No, No.

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{16}$

+ $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$: $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{4}{16}$

+ $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$: $\left(\frac{1}{4}\right) \left(\frac{1}{4}\right) = \frac{1}{16}$

Warmup

A spinner is chosen and spun twice, resulting in No, No.

$$+ P(\text{No}, \text{No}) \text{ if } \pi = \frac{1}{4}: \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{16}$$

$$+ P(\text{No}, \text{No}) \text{ if } \pi = \frac{1}{2}: \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{4}{16}$$

$$+ P(\text{No}, \text{No}) \text{ if } \pi = \frac{3}{4}: \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) = \frac{1}{16}$$

Which spinner would you guess was spun?

Warmup

A spinner is chosen and spun twice, resulting in No, No.

$$+ P(\text{No}, \text{No}) \text{ if } \pi = \frac{1}{4}: \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{16}$$

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$$+ P(\text{No}, \text{No}) \text{ if } \pi = \frac{3}{4}: \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) = \frac{1}{16}$$

Which spinner would you guess was spun? **Spinner A, because the probability of the observed data is the highest**

Warmup

A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{4}$:

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{2}$:

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{3}{4}$:

Warmup

A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{4}$: $\frac{9}{256}$

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{2}$: $\frac{16}{256}$

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{3}{4}$: $\frac{9}{256}$

Which spinner would you guess was spun?

Warmup

A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{4}$: $\frac{9}{256}$

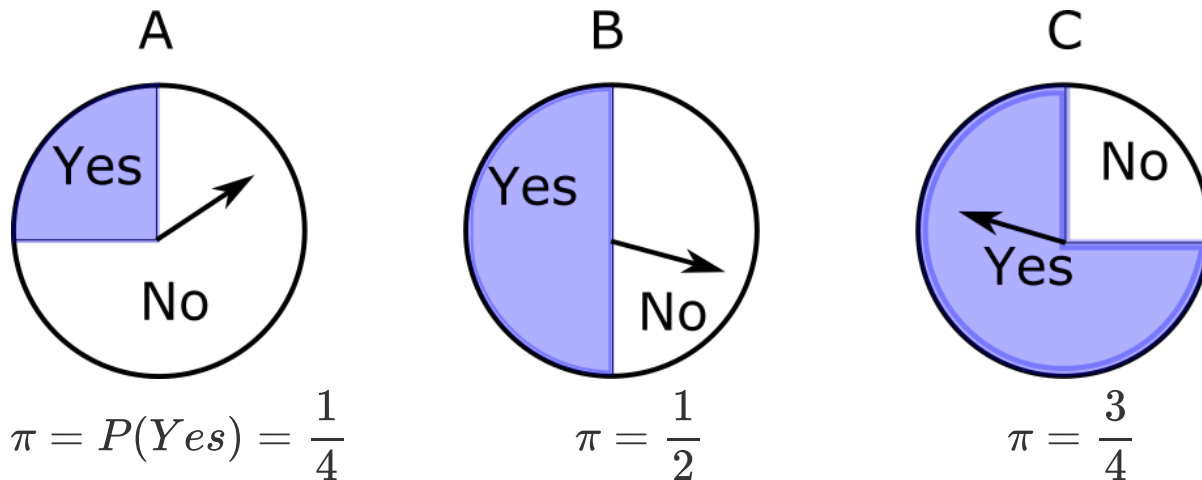
+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{2}$: $\frac{16}{256}$

+ $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{3}{4}$: $\frac{9}{256}$

Which spinner would you guess was spun? **Spinner B, because the probability of the observed data is the highest**

Likelihood

Likelihood: the probability of the observed data, regarded as a function of the parameters



$\pi = \text{parameter value}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$P(\text{No}, \text{No}) = \text{likelihood} = L(\pi)$	$\frac{9}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

$\pi = \text{parameter value}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$P(\text{No}, \text{No}) = \text{likelihood} = L(\pi)$	$\frac{9}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Maximum likelihood estimate: $\hat{\pi} = \frac{1}{4}$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

Steps for maximum likelihood estimation:

- + *Likelihood:* For each value of the parameter, compute the probability of the observed data, $P(\text{data})$
- + *Maximize:* Find the parameter value that gives the largest $P(\text{data})$

Maximum likelihood for logistic regression

Putting data: Putt at lengths 3, 4, 5, 6, and 7 feet. Record whether you make the put ($y = 1$) or miss ($y = 0$). Let $\pi = P(y = 1)$.

Regression model:

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{ length} \quad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

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Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3$, $\beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

Maximum likelihood for logistic regression

Regression model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length} \quad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3$, $\beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

Length	3	4	5	6	7
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

Maximum likelihood for logistic regression

Regression model:

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{ length} \quad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3$, $\beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

Length	3	4	5	6	7
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.818)(0.818)(0.731)(1 - 0.622)(1 - 0.378) = 0.115$$

Class activity

Work on the class activity (handout).

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 3$ and $\beta_1 = -0.5$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7	
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}}$	0.818	0.731	0.622	0.5	0.378

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.818)^3 (0.731)^2 (0.622)^3 (1 - 0.622)^2 (0.5) (1 - 0.5) \\ \cdot (0.378) (1 - 0.378)^2$$

$$= 0.000368$$

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 4$ and $\beta_1 = -0.75$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7	
π	$\frac{\exp\{4 - 0.75(3)\}}{1 + \exp\{4 - 0.75(3)\}}$	0.852	0.731	0.562	0.378	0.223

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.852)^3 (0.731)^2 (0.562)^3 (1 - 0.562)^2 (0.378) (1 - 0.378) \\ \cdot (0.223) (1 - 0.223)^2$$

$$= 0.000356$$

Maximum likelihood for logistic regression

Likelihood:

- + For estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, $\hat{\pi} = \frac{\exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}{1 + \exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}$
- + $L(\hat{\beta}_0, \hat{\beta}_1) = P(\text{data})$

Maximize:

- + Choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

In the class activity examples, we have only considered a few different possibilities for $\hat{\beta}_0$ and $\hat{\beta}_1$. In reality, we consider all real numbers as possible values.

Fitting logistic regression

Logistic regression with maximum likelihood

- + **Model:**

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 x$$

- + **Measure of fit:**

$L(\hat{\beta}_0, \hat{\beta}_1)$ = probability of observed data for estimates $\hat{\beta}_0, \hat{\beta}_1$

- + **Optimize:** choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

Linear regression with least squares

- + **Model:** $y = \beta_0 + \beta_1 x + \varepsilon$

- + **Measure of fit:**

$$SSE = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- + **Optimize:** choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize SSE